Quantification: 701_z19c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 19 |
| interior lacuna count | 19 |
| frame-edge lacuna count | 0 |

Interior lacunae only: mean, median and sample SD (n = 19)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1802.47 | 1055 | 1961.74 | px^2 |
| major_axis_length_px | lacuna shape | 67.9675 | 66.6467 | 29.4363 | px |
| minor_axis_length_px | lacuna shape | 34.9677 | 28.9914 | 19.8195 | px |
| aspect_ratio | lacuna shape | 2.1144 | 2.1401 | 0.7337 | unitless |
| eccentricity | lacuna shape | 0.8061 | 0.8841 | 0.182 | unitless |
| solidity | lacuna shape | 0.7624 | 0.7238 | 0.089 | unitless |
| roots_count | headline | 9.6316 | 8 | 5.5097 | count |
| ring_length_r30_px | headline | 375.842 | 321 | 180.157 | px |
| ring_length_r60_px | reported | 782.842 | 654 | 377.46 | px |
| owned_length_px | ownership-dependent | 1014.43 | 770.517 | 771.546 | px |
| edge_count | ownership-dependent | 65.0526 | 49 | 49.6437 | count |
| mean_edge_length_px | ownership-dependent | 15.7472 | 15.286 | 2.5409 | px |
| perimeter_px | normalised | 247.023 | 199.622 | 209.483 | px |
| ring_area_r30_px2 | normalised | 6970.11 | 6194 | 2837.67 | px^2 |
| ring_area_r60_px2 | normalised | 15425.6 | 14585 | 6029.06 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0 | unitless |
| in_frame_fraction_r60 | normalised | 0.9957 | 1 | 0.0158 | unitless |
| ring_density_r30 | normalised | 0.053625 | 0.0531361 | 0.010021 | px^-1 |
| ring_density_r60 | normalised | 0.049899 | 0.0490588 | 0.0095752 | px^-1 |
| roots_per_100px_perimeter | normalised | 4.623 | 4.2973 | 2.2845 | per 100 px |
| ring_attached_length_r30_px | network v2 | 330.789 | 279 | 186.279 | px |
| ring_attached_length_r60_px | network v2 | 596.684 | 484 | 374.458 | px |
| ring_length_w_r30_px | network v2 | 422.867 | 355.279 | 200.766 | px |
| ring_length_w_r60_px | network v2 | 880.072 | 743.37 | 423.764 | px |
| sholl_crossings_r10 | network v2 | 9.1579 | 7 | 4.7171 | count |
| sholl_crossings_r20 | network v2 | 9.4211 | 8 | 4.3117 | count |
| sholl_crossings_r30 | network v2 | 9.7895 | 9 | 4.7209 | count |
| loop_count | loops, report only | 3.5789 | 1 | 5.3262 | count |
| loop_edge_share | loops, report only | 0.1491 | 0.1146 | 0.1348 | share |
| ring_width_mean_r30_px | width | 5.8265 | 6.0006 | 1.6125 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 787838 | px^2 |
| total_skeleton_length_px | 34738 | px |
| canalicular_length_density_per_px | 0.0440928 | px^-1 |
| junction_count | 922 | count |
| junction_density_per_px2 | 0.00117029 | px^-2 |
| lacuna_count | 19 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 2.412e-05 | px^-2 |
| skeleton_component_count | 583 | count |
| mean_component_length_px | 59.5849 | px |
| median_component_length_px | 13 | px |
| field_length_density_w_per_px | 0.0495589 | px^-1 |
| field_density_without_flagged_per_px | 0.0290095 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 379 | px |
| unattached_component_count | 20 | count |
| canalicular_length_density_attached_per_px | 0.0440928 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0445739 | px^-1 |
| excluded_area_px2 | 232055 | px^2 |
| analysed_field_area_px2 | 816521 | px^2 |
| loop_count | 120 | count |
| loop_edge_share | 0.1764 | share |
| edge_count | 3016 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 6.5804 | px |
| width_median_px | 6.3246 | px |
| width_p10_px | 2.8284 | px |
| width_p90_px | 10 | px |
| width_pixels_used | 22823 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 653 | 48.9306 | 18.0087 | 2.7171 | 0.9298 | 0.8718 | 1.3396 | 244.608 | 191.13 | False |
| 2 | 439 | 24.8605 | 23.4655 | 1.0594 | 0.3303 | 0.8905 | -1.2149 | 286.371 | 223.007 | False |
| 3 | 511 | 42.2275 | 16.9827 | 2.4865 | 0.9156 | 0.8705 | 0.9114 | 308.108 | 170.11 | False |
| 4 | 663 | 44.6399 | 20.5821 | 2.1689 | 0.8874 | 0.8511 | 1.5097 | 320.038 | 647.3 | False |
| 5 | 1055 | 56.5297 | 24.2615 | 2.33 | 0.9032 | 0.8843 | 1.0383 | 419.101 | 230.957 | False |
| 6 | 1893 | 74.136 | 39.3577 | 1.8836 | 0.8474 | 0.6949 | -1.3345 | 688.738 | 268.372 | False |
| 7 | 1441 | 68.9456 | 33.0049 | 2.089 | 0.878 | 0.7238 | -1.5587 | 747.817 | 126.852 | False |
| 8 | 3491 | 77.9644 | 65.3058 | 1.1938 | 0.5462 | 0.7932 | -1.5218 | 773.022 | 646.082 | False |
| 9 | 988 | 66.6467 | 25.2664 | 2.6378 | 0.9254 | 0.6694 | -1.3153 | 754.104 | 873.76 | False |
| 10 | 8184 | 138.719 | 91.3085 | 1.5192 | 0.7528 | 0.6732 | -1.4216 | 853.859 | 480.226 | False |
| 11 | 1270 | 71.948 | 33.6183 | 2.1401 | 0.8841 | 0.6499 | -1.4065 | 891.32 | 930.428 | False |
| 12 | 1577 | 97.9 | 24.8484 | 3.9399 | 0.9673 | 0.7152 | -1.2598 | 906.939 | 130.226 | False |
| 13 | 957 | 43.0583 | 35.4718 | 1.2139 | 0.5669 | 0.6965 | 1.5597 | 917.457 | 535.305 | False |
| 14 | 1036 | 53.8662 | 28.9914 | 1.858 | 0.8428 | 0.7623 | -1.4483 | 935.929 | 453.511 | False |
| 15 | 2879 | 109.14 | 44.0418 | 2.4781 | 0.915 | 0.6894 | -1.4893 | 959.511 | 178.398 | False |
| 16 | | | | | | | | 361.992 | 272.702 | False |
| 17 | | | | | | | | 528.416 | 314.222 | False |
| 18 | | | | | | | | 549.191 | 341.457 | False |
| 19 | | | | | | | | 659.583 | 359.618 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 4 | 206 | 443 | 367.22 | 18 | 20.4011 | 3.2714 |
| 2 | 6 | 194 | 252 | 206.61 | 19 | 10.8742 | 5.4004 |
| 3 | 3 | 216 | 385 | 430.889 | 21 | 20.5185 | 6.3787 |
| 4 | 5 | 309 | 716 | 801.826 | 54 | 14.8486 | 6.0006 |
| 5 | 7 | 270 | 573 | 685.985 | 39 | 17.5894 | 5.6034 |
| 6 | 9 | 427 | 933 | 1214.83 | 88 | 13.8049 | 5.6729 |
| 7 | 8 | 403 | 1011 | 1181.72 | 66 | 17.9048 | 6.6716 |
| 8 | 16 | 645 | 1412 | 3048.61 | 189 | 16.1302 | 7.0709 |
| 9 | 8 | 350 | 822 | 497.86 | 31 | 16.06 | 6.083 |
| 10 | 25 | 853 | 1634 | 2329.87 | 166 | 14.0354 | 8.4678 |
| 11 | 12 | 409 | 1090 | 1869.23 | 96 | 19.4712 | 5.9898 |
| 12 | 16 | 537 | 1102 | 1467.46 | 96 | 15.286 | 7.7821 |
| 13 | 11 | 322 | 589 | 664.215 | 49 | 13.5554 | 2.8284 |
| 14 | 10 | 303 | 654 | 1290.84 | 88 | 14.6687 | 4.5701 |
| 15 | 17 | 655 | 1217 | 1514.27 | 102 | 14.8458 | 7.4838 |
| 16 | 7 | 189 | 392 | 205.723 | 14 | 14.6945 | 2.4142 |
| 17 | 6 | 321 | 543 | 376.836 | 30 | 12.5612 | 5.913 |
| 18 | 5 | 237 | 485 | 349.664 | 22 | 15.8938 | 6.0423 |
| 19 | 8 | 295 | 621 | 770.517 | 48 | 16.0524 | 7.0591 |

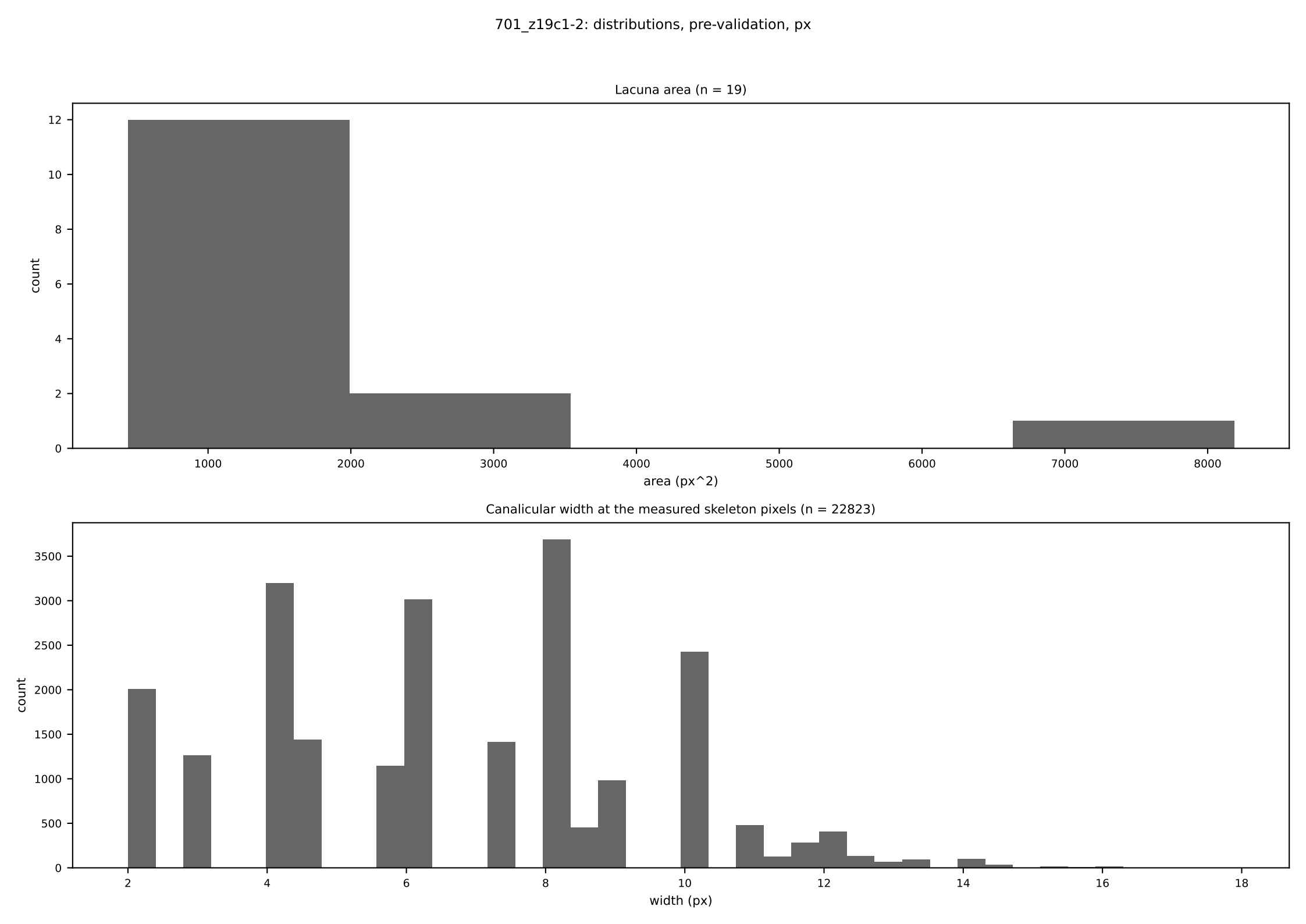
Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 122.225 | 4947 | 11212 | 1 | 1 | 0.0416414 | 0.0395112 | 3.2726 |
| 2 | 94.7107 | 3651 | 5943 | 1 | 1 | 0.0531361 | 0.0424028 | 6.3351 |
| 3 | 114.711 | 4934 | 9822 | 1 | 1 | 0.0437779 | 0.0391977 | 2.6153 |
| 4 | 134.61 | 6194 | 17999 | 1 | 1 | 0.049887 | 0.03978 | 3.7144 |
| 5 | 162.894 | 6766 | 14585 | 1 | 1 | 0.0399054 | 0.0392869 | 4.2973 |
| 6 | 260.777 | 8904 | 19018 | 1 | 1 | 0.047956 | 0.0490588 | 3.4512 |
| 7 | 308.818 | 8500 | 21039 | 1 | 1 | 0.0474118 | 0.0480536 | 2.5905 |
| 8 | 337.446 | 10732 | 24742 | 1 | 1 | 0.0601006 | 0.057069 | 4.7415 |
| 9 | 200.25 | 7746 | 20901 | 1 | 1 | 0.0451846 | 0.0393283 | 3.995 |
| 10 | 987.088 | 14681 | 26883 | 1 | 1 | 0.0581023 | 0.0607819 | 2.5327 |
| 11 | 267.984 | 8463 | 21768 | 1 | 0.9865 | 0.048328 | 0.0500735 | 4.4779 |
| 12 | 301.404 | 8091 | 17268 | 1 | 1 | 0.06637 | 0.0638175 | 5.3085 |
| 13 | 255.806 | 4850 | 9732 | 1 | 1 | 0.0663917 | 0.060522 | 4.3001 |
| 14 | 199.622 | 5174 | 11017 | 1 | 1 | 0.058562 | 0.0593628 | 5.0095 |
| 15 | 503.269 | 10225 | 19270 | 1 | 0.9316 | 0.0640587 | 0.0631552 | 3.3779 |
| 16 | 97.8823 | 5235 | 10032 | 1 | 1 | 0.0361032 | 0.039075 | 7.1514 |
| 17 | 169.38 | 5166 | 12626 | 1 | 1 | 0.062137 | 0.0430065 | 3.5423 |
| 18 | 111.196 | 3920 | 9020 | 1 | 1 | 0.0604592 | 0.0537694 | 4.4966 |
| 19 | 63.3553 | 4253 | 10209 | 1 | 1 | 0.0693628 | 0.0608287 | 12.6272 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 193 | 302 | 234.238 | 494.718 | 5 | 5 | 5 |
| 2 | 148 | 152 | 217.51 | 280.38 | 6 | 5 | 2 |
| 3 | 147 | 267 | 250.693 | 448.274 | 5 | 8 | 7 |
| 4 | 251 | 359 | 343.007 | 802.754 | 9 | 7 | 9 |
| 5 | 237 | 382 | 312.392 | 638.671 | 7 | 7 | 7 |
| 6 | 289 | 661 | 478.333 | 1061.06 | 7 | 16 | 16 |
| 7 | 333 | 747 | 448.363 | 1126.75 | 10 | 10 | 13 |
| 8 | 619 | 1136 | 719.487 | 1576.64 | 17 | 17 | 14 |
| 9 | 243 | 461 | 401.605 | 931.708 | 6 | 12 | 11 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 10 | 832 | 1544 | 958.968 | 1842.09 | 19 | 17 | 17 |
| 11 | 328 | 715 | 457.433 | 1229.84 | 9 | 10 | 15 |
| 12 | 533 | 1062 | 599.404 | 1237.85 | 19 | 13 | 10 |
| 13 | 315 | 484 | 363.907 | 663.286 | 7 | 8 | 6 |
| 14 | 279 | 597 | 343.321 | 743.37 | 9 | 4 | 6 |
| 15 | 621 | 1079 | 735.186 | 1364.77 | 15 | 14 | 17 |
| 16 | 159 | 162 | 214.309 | 442.889 | 6 | 4 | 4 |
| 17 | 302 | 347 | 355.279 | 604.659 | 6 | 9 | 6 |
| 18 | 229 | 344 | 266.823 | 548.86 | 7 | 5 | 7 |
| 19 | 227 | 536 | 334.208 | 682.789 | 5 | 8 | 14 |


701_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22823)
count
width (px)